Restoring natural fire and the cost of suppression in American Wilderness: Evidence from a policy experiment

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Disclaimer

This dissertation overlaps with the summative research project submitted for GY 460 (14582, 2019; word count: 4,996). This paper is based on the same policy experiment and part of this paper's analysis shares an identification strategy with 14582 (2019).

In addition to explaining the ecological underpinnings more fully, substantially expanding the literature review, and providing a conceptual framework within dynamic optimisation, I build on 14582 (2019) by overhauling the methodology and data sources. 14582 (2019) used data aggregated by year and administrative unit. This dissertation identifies treatment effects with a higher degree of precision using daily weather data, incident-level fire data, and geospatial Landsat fire perimeter data to explicitly map overlap between fires (using mixed effects rather than fixed effects). This paper uses seven additional years of data and two new instrumental variables, assesses an instrumental variable specification based on thresholds in safe burning conditions, and models decay in treatment effects over time. Data limitations prevent an extension of the geographic study area (Section 6). Finally, I extend 14582 (2019) by investigating the mechanism by which fuel treatment affects suppression cost by estimating the effect of treatment on fire intensity and fuel availability (necessitating the use of two additional models and an additional high-resolution satellite dataset).

Words: 200

1 Introduction

Wildfires in the American West are becoming increasingly destructive and difficult to contain (Westerling et al., 2006). Fire management costs and damages have risen steadily, with suppression expenditures alone topping $3 billion in 2018 (NIFC, 2018). These rising costs, along with a succession of devastating fires in California, have focused attention on long-standing forest management practices. In the wake of the devastating fires of 1910, land management agencies adopted a policy of total fire suppression, also known as the 10 A.M. policy (as all fires were to be put out by 10 A.M. the day after they were discovered; Chung, 2015). By depriving landscapes of natural fire, this policy has led to an accumulation of overgrowth that has fuelled increasingly severe fires. That effective suppression has led to worsening fires is known as the 'wildfire paradox' (Ager et al., 2014; Charnley et al., 2015).

After 60 years of all-out suppression, the importance of regular forest fire in healthy ecosystems was becoming clear. This realisation prompted a paradigmatic shift in fire management (Kilgore, 2007; Miller, 2002; Van Wagtendonk, 2007). To restore natural conditions, agencies began to undertake measures to reduce the build-up of overgrowth resulting from suppression (through the intentional lighting of fires or manual clearing of fuels). This paper looks at the impacts of one such fuel treatment policy known as wildland fire use (WFU).

Under WFU, naturally ignited fires are not suppressed; they are monitored and allowed to burn as fuel treatment. By allowing natural fires to run their course, it was hoped ecosystems would revert to their natural fire regimes, reducing the incidence of severe fires and lowering suppression costs. The policy has been described as a "natural fire management experiment," but has received little attention from economists (Miller, 2014, p. 20). This paper estimates the impact of wildland fire use on the suppression cost of successive fires using weather conditions and ignition cause as instrumental variables for treatment assignment. This paper builds on the existing literature by examining the mechanism by which WFU affects suppression cost by estimating the impact of treatment on both fire intensity and fuel availability.

I find that wildland fire use (planned treatment) reduced total and per-hectare sup-

pression costs, although this effect is statistically insignificant. Wildfire on the other hand (fires that are not managed for fuel reduction but escape or resist control; i.e., unplanned treatment), significantly reduced suppression costs, suggesting that if applied on a larger scale, WFU could achieve similar reductions. Impacts on fire intensity and fuel availability are inconclusive.

This paper proceeds as follows. In Section 2, I give a brief history of the policy and its context. Section 3 gives the conceptual framework; Section 4 presents a review of relevant literature. Sections 5 and 6 respectively describe the research design and sample characteristics, followed by the empirical model in Section 7. Section 8 presents results and a discussion. The paper closes with concluding remarks in Section 9.

2 Background

2.1 Wilderness fire management

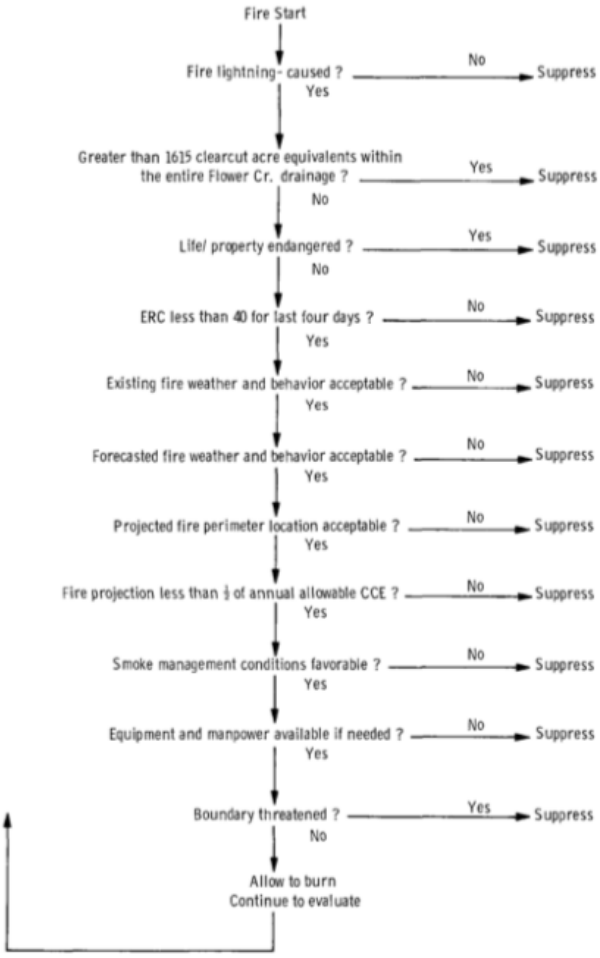
Some 800 designated Wilderness Areas comprise the National Wilderness Preservation System (NWPS), which is the setting of this study. Wilderness Areas are large, undisturbed, usually remote public lands that are strictly protected and maintained in their natural condition. The bulk of WFU treatment has been applied in Wilderness Areas, where it may be the most feasible option due to these areas' remote character, unique legal constraints, and the public's demand for low-impact management (Miller, 2002). Wilderness Area restrictions make fuel treatments more "complicated, expensive, and controversial" and mechanical fuel treatments like selective logging are seen as unacceptable (Charnley et al., 2015 p. 334; Parsons et al., 2003).

The Forest Service and National Park Service began applying WFU experimentally in the 1960s, but federal guidance was not issued until 1995 (Miller, 2002). WFU gained traction in the interim, but progress was halted in 1988 when fires being managed for fuel reduction benefits escaped and burned more than half a million hectares in and around Yellowstone National Park (Van Wagtendonk, 2007). In 1995, lawmakers issued the Fire Management Policy and Program Review (Miller, 2002). Van Wagtendonk (2007) notes

that within several years, the 1995 Review had revived WFU programmes. [1]

There are significant challenges associated with implementing wildland fire use. If a landscape has been deprived of forest fire for 100-odd years, the reintroduction of fire can do more harm than good (due to the ecosystem's weakened resilience). This has been addressed in much greater depth by Landres et al. (2000), Parsons et al. (2003), and Parsons & Landres (1998). Mercer et al. (2007) outlines the practical obstacles to implementing WFU: air quality concerns, public opposition, regulatory restrictions, danger to both human and ecological values-at-risk, risk of escaped fires, and lack of funds and incentives.

Figure 1: Flower Creek management unit decision scheme [2]



[1] The policy was updated again in 2001 and 2009.

[2] Schulte & Davis (1980) as cited in Fischer (1984). A decision scheme from Kootenai National Forest is also cited by 14582 (2019). Decision schemes vary by Wilderness Area, as each adopts its own fire management plan.

Figure 1 demonstrates the risk management conditions that must be in place in order for WFU to be applied and explains why most ecologically significant Wilderness fires are still suppressed (Parsons & Landres, 1998; Miller, 2002). These conditions only determine which fires are *eligible* for WFU. Whether to let burn or suppress still comes down to individual judgement. Van Wagtendonk (1995) notes that land managers fear the risk to their careers and reputations in allowing a fire to burn and Charnley et al. (2015) finds that social acceptability is a key factor in decision-making.

2.2 Ecological foundations

In order to burn, fire needs heat, oxygen and fuel. Fuel availability is determined by the type and amount of vegetation, its continuity, and its moisture content. High fuel loads and continuity, along with low moisture, are generally thought to contribute to fire intensity.

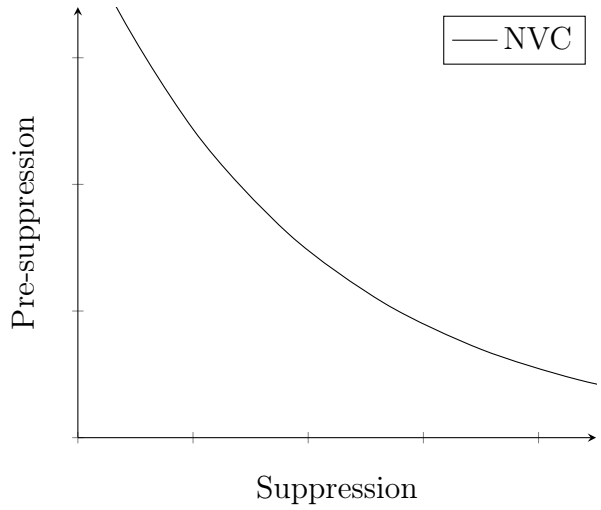
Recurrent fires create a naturally patchy landscape with breaks in vegetation to contain fire spread (Charnley et al., 2015; Fernandes & Botelho, 2003). Fire exclusion has led to large, contiguous forests that allow fires that are not immediately contained to go unchecked. The effects of fire exclusion policy have been most pronounced in forests with low fire return intervals (i.e., forests that naturally experience frequent burns). It so happens that these forests dominate the Western landscape (e.g., long-needle pine species like Ponderosa Pine; Williams, 1995).

Fuel treatment is believed to reduce future fire intensity by consuming flammable vegetation and thereby reducing fuel availability (Cochrane et al., 2012; Prestemon et al., 2002). However, as noted by Agee & Skinner (2005), the true relationship between fuel treatment and fire behaviour is not quite so straightforward. Fuel treatments may also expose the forest understory to higher temperatures and wind speeds, *exacerbating* fire intensity. Because of these competing effects, Cochrane et al. (2012) describe the effect of fuel treatments as "redistributing" (rather than reducing) fire risk (p. 357).

3 Conceptual framework

The intertemporal dynamics of fire management make it well suited to dynamic control theory. Fire managers smooth costs over time by scheduling fuel treatments and other management tools over the management horizon, minimising the net present value of social cost (Mercer et al., 2007). This is formalised in the cost-plus-net-value-change ($C{+}NVC$) model, which represents the total costs of wildfire (Donovan & Rideout, 2003; Hesseln, 2000). The cost term (C) comprises pre-suppression and suppression costs and the net value change function (NVC) represents damages net of benefits arising from wildfire. Pre-suppression costs are capital investments made at the start of a fire season; suppression costs are incurred over the course of the season (Donovan & Rideout, 2003). The former include fuel treatment and equipment purchases while the latter include expenses such as firefighter wages, fire retardant, and petrol. Fitch et al. (2018) note that pre-suppression and suppression costs are related through the NVC function, the general form of which is illustrated below and attributed to Donovan & Rideout (2003; p. 319). A mathematical description of the model is offered by Thomas et al. (2017).

Figure 2: NVC function (Donovan & Rideout, 2003)



Higher levels of fuel treatment (pre-suppression) are associated with lower levels of suppression. This study tests whether this holds in practice.

4 Literature review

4.1 Fuel treatment evaluation

Determining optimal levels of fuel treatment requires the specification of a wildfire production function and fire simulation modelling to compare alternative policy regimes (Mercer et al., 2007; Prestemon et al., 2002). Mercer et al. (2007) undertake such an analysis at the county level in Florida and find prescribed burning at a rate of 41,000 to 48,000 acres per year to be optimal. In addition to determining the optimal level of fuel treatment, a burgeoning body of literature is taking a more nuanced approach by investigating the optimal siting and combinations of treatments (Prestemon et al., 2002).

A related body of literature seeks not to determine optimal levels of fuel treatment, but simply to estimate the relationship between fuel treatment and subsequent fire behaviour, costs, or risk. Simulation modelling has also informed the fuel treatment evaluation literature. Using fire behaviour models, researchers can input observed weather and topology and simulate fire sequences under different policies, essentially creating counterfactuals (Cochrane et al., 2012; Fitch et al., 2018). The focus of this paper, however, is the empirical findings.

Mercer et al. (2007) present the empirical results that are used as inputs to their simulation model. Using county-level panel data from Florida, they report that each percentage increase in prescribed burn reduces intensity-weighted risk 0.71% over a two-year span and 0.27% over a three-year span. The effects of prescribed burning were similar to those of previous wildfire. Prestemon et al. (2002) undertake a similar analysis, using county-level panel data from Florida to estimate the effect of prescribed fire (planned treatment) and wildfire (unplanned treatment) on wildfire risk (probability of ignition). They report that previous wildfire has a significant and robust suppressive effect on subsequent fire risk, but the results are mixed for fuel treatment.

With the availability and capability of geospatial processing tools expanding in recent years, there has been a shift toward incident-level rather than county-level analyses. Loomis et al. (2019) undertake one of the few existing large-scale, incident-level analyses of prescribed burning and mechanical thinning on wildfire suppression cost and wildfire

property damages. Using U.S. Forest Service data, they estimate a simple linear regression controlling for a number of geographic variables. Overall, fuel treatment did not significantly reduce suppression costs (except in California) but did reduce the number of structures damaged by wildfire.

Within the empirical fuel treatment evaluation literature, there has been little demonstration of advanced methods. The exception is Butry (2005) and Butry (2009) who use instrumental variables (IV) and propensity score matching (PSM) to account for selection bias in the application of prescribed burning. The PSM approach (Butry, 2005; Butry, 2009) matches treated and untreated fires according to their probability of treatment based on observable characteristics. Overall, a 1% increase in prescribed burn area is associated with a statistically insignificant 0.0138% decrease in wildfire intensity-weighted acres.

To isolate the random component of prescribed fire application, Butry (2005) uses a vector of instrumental variables that includes past weather conditions, landscape characteristics, distances to populations at risk, and neighbouring prescribed fire. The reasoning behind these instruments is that prescribed fire is less likely to be carried out under severe weather conditions, or where smoke from treatment fires could harm children or the elderly. Butry (2005) reports statistically insignificant reductions in wildfire acres and flame length. However, there is reason to doubt the validity of Butry's instruments. If instruments affect the outcome through channels other than the instrumented variable, they are said to be endogenous. Several of the instruments in his estimation likely affect suppression cost through channels other than their effect on treatment assignment (e.g., weather conditions with persistent effects and fixed geographic or landscape characteristics).

This paper also considers the causal pathway by which fuel treatment affects suppression cost by estimating treatment effects on fire intensity and vegetation condition. Several of the papers already described estimate treatment effects on fire intensity as the dependent variable (Mercer et al., 2007; Butry, 2005; 2009). However, studies that consider vegetation condition or fire regime are less common. Vaillant & Reinhardt (2017) compare fuel treatment and historic fire occurrence to determine whether fuel treatment "compensates for the disturbance deficit caused by fire suppression" (p. 300). They report

that fuel treatment is not applied on a wide enough scale to make up for natural fire.

4.2 Wildland fire use

WFU is carried out on a much smaller scale relative to traditional prescribed burning. By virtue of their isolation, Wilderness Area fires are less expensive and less damaging than typical forest fires, making Wilderness fire management policy less publicly visible (Hand et al., 2016). For example, Hand et al. (2016) and Loomis et al. (2019) both address Wilderness fire only to justify why they ignore Wilderness fire.

There are "scant few" studies of wildland fire use in the social sciences (Miller, 2014, p. 22), and fewer still in economics. The exception is Houtman et al. (2013), who simulate fire sequences over a small area in the Deschutes National Forest to estimate the effect of the policy on the present value of future reductions in suppression costs. They find the present value of suppression cost savings to be a statistically significant $34 per hectare.

The ecological literature is more fruitful but even the ecological literature is limited to small-scale case studies (Parks et al., 2015). Such studies tend to compare ecological measures and patterns of fire behaviour between areas of spatial overlap within a single forest or designated area (Miller, 2014). Overlapping fuel treatments create mosaics that act as containment lines (fuel breaks). This has been observed in Yellowstone and Yosemite National Park (Van Wagtendonk, 1995; Miller, 2002). Parks et al. (2015) offers perhaps the most comprehensive of these studies; investigating the efficacy of previous fire as fuel breaks in successive fires in four Wilderness Areas. They do not explicitly account for fuel treatment; rather, they use previous wildfire in general as a proxy for WFU. They find that fire "clearly" limits the progression of subsequent fires (p. 1489). This effect varies regionally and decays over time.

4.3 Gaps in the literature

Overall, there is a noted shortage of rigorous analyses of fuel treatment effectiveness (Mercer et al., 2008; Hesseln, 2000). This is particularly pronounced when it comes to empirical estimates of fuel treatment efficacy on a broad geographic scale and with the exception of Butry (2005; 2009), there has been little application of advanced empirical

methods.

Existing studies are subject to important weaknesses. Loomis et al. (2019) offer one of the most comprehensive empirical analyses of prescribed burning in the U.S. but do not control for weather conditions, model the persistence of the treatment effect, or rigorously account for selection bias. Butry's PSM and IV analyses have been applied on a small geographic scale, and are prone to endogeneity.

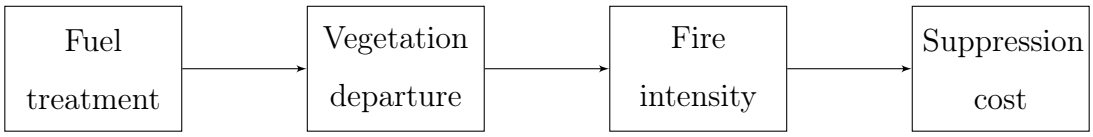
Finally, the policy in question has received very little treatment from economic researchers. The economic literature is focused on wildfires that occur in the wildland urban interface (WUI), where the bulk of suppression expenditures are incurred. However, designated Wilderness Areas cover an area larger than the state of California and are some of the most important conservation sites in the country. How they are managed should be of greater interest to economists.

This paper builds on and expands the existing literature. Like Loomis et al. (2019), this study considers incident-level fire costs with precise spatial identification of treatment effects, but with the inclusion of daily weather data and an instrumental variable strategy. It builds on Butry's (2005) IV analysis in its choice of instrumental variables, and extends Parks et al. (2015) by widening the study area and considering economic outcomes. Finally, this paper goes further than all the above papers by investigating the mechanism by which fuel treatment is thought to occur.

5 Research design and identification strategy

I consider the effect of prior WFU on three different dependent variables: suppression cost, fire intensity, and vegetation departure, which are related via the schematic below.

Figure 3: Chain of causality



Fire suppression cost and intensity are both characteristics of individual fires. Because each fire only occurs once, suppression and intensity data constitute repeated cross-

sectional data. For these regressions, the treatment variable is defined as the cumulative number of hectares of fuel treatment within a given wildfire's burn perimeter (following Loomis et al., 2019; Butry, 2005; Butry, 2009). Characteristics of previous overlapped fires are used as instrumental variables to account for endogeneity.

Vegetation departure is a characteristic of the landscape, so I calculate mean vegetation departure indices within each fire perimeter across multiple years. Rather than defining treatment based on spatial overlap, a difference-in-difference model compares vegetation departure between burned and unburned areas and before and after the occurrence of fire.

5.1 Instrumental variables

Traditional prescribed burning, in which land managers intentionally light fires for fuel reduction benefits, is a common fuel management strategy that affords land managers discretion in the siting and timing of burns. What distinguishes WFU from prescribed burning is the lack of control granted to the land manager (14582, 2019). Instead of lighting treatment fires, managers are faced with unplanned ignitions which they assign to a treatment policy (suppress or allow to burn with continuing evaluation) based on existing and forecasted conditions (Figure 1).

To identify treatment effects on suppression cost and intensity, this analysis exploits the random variation in source of ignition (treatment eligibility) and in the transient weather conditions that influence treatment assignment. Stock & Watson (2015) describe two types of quasi-experiments: one where treatment is randomly assigned and one where treatment is only partially randomly assigned. WFU falls into the second category. Although land managers have less control over the siting and timing of treatment than in traditional management-ignited prescribed fire, they are clearly involved in the treatment assignment process. This introduces selection bias.

In addition to selection bias, simultaneity also threatens the internal validity of fuel treatment evaluation (Butry, 2005). Simultaneity refers to feedback between the independent and dependent variables in a model. In the suppression cost estimation, the lagged nature of the treatment variable (cumulative *past* fire) reduces the threat of simultaneity.

However, in the intensity model, I follow Butry (2009) and Mercer et al. (2007), both of whom use intensity-weighted area burned as a dependent variable, which is cited as a better measure of fire behaviour and damage than intensity or fire size alone. However, fire size is likely subject to simultaneity. Fuel treatment may affect fire size, but fire size may affect the probability a fire is exposed to previous treatment (a larger fire has a higher spatial probability of overlapping a previous burn).

To account for both sources of endogeneity, this analysis uses the source of ignition, as well as a set of transitory weather conditions, as instrumental variables. An instrumental variable Z describes the random variation in the endogenous variable WFU. Therefore, the instrument must be correlated with treatment assignment and uncorrelated with the unobservable determinants of suppression cost. Formally, these conditions are expressed as:

1. Relevance: $corr(Z, WFU) \neq 0$

2. Exogeneity: $corr(Z, u) = 0$

To satisfy the exogeneity condition, an instrumental variable Z should only affect suppression cost through its effect on treatment assignment WFU. This invalidates a number of the instrumental variables employed in Butry's (2005) analysis, including weather variables with intertemporal effects (precipitation) and fixed terrain characteristics (distance to hospitals).

5.1.1 Ignition source

The first instrument is the type of ignition; i.e., whether the fire was ignited naturally (by lightning) or not. [3] Only natural fires can be managed for fuel reduction benefits; anthropogenically caused fires (by accident or arson) are suppressed. Because ignition source determines treatment eligibility, it is reasonable to expect it to be correlated with treatment assignment.

Lightning has not been applied as an instrumental variable, but it has been addressed in the wildland fire modelling literature, and its use exposes the vulnerabilities of this

[3]Fires may also be naturally ignited by lava, but there are no lava-ignited fires in this sample.

approach. Although lightning is the result of atmospheric processes, and would theoretically be considered exogenous, the story becomes somewhat more complicated when you consider the spatial distribution of fire ignitions by type. Human-caused fires are more likely in areas that are accessible or heavily travelled. Fires ignited in these areas may be fought more aggressively to protect human values-at-risk. Alternatively, it could cost less to fight these fires due to easier access to fire lines and fuel breaks.

However, these concerns do not invalidate ignition type as an IV for a number of reasons. First is the nature of the study setting. All activity but low-impact recreation is prohibited in Wilderness Areas, reducing the potential for concentrations in human activity (not even roads are allowed). Second, the model includes controls for distance from values-at-risk and fuel breaks, including distance from Wilderness Area boundary, priority watersheds, populated areas, major roads, and bodies of water. Finally, Prestemon et al. (2002) modelled the suppressive effect of fire by ignition source and the results were broadly similar across lightning, arson, and accident-ignited fires.

5.1.2 Weather: wind speed, air temperature, relative humidity

Butry's (2005) evaluation of prescribed fire used relative humidity, wind speed, temperature, and precipitation as instruments. The intuition for these is that treatment is less likely to be carried out under weather conditions that contribute to extreme fire behaviour (high temperatures, low moisture, high wind speeds). Letting a fire burn under such conditions increases the chances that the fire escapes and causes damage to human or ecological values.

Weather, like lightning, is the result of atmospheric processes that can usually be considered exogenous. However, exogeneity is violated if weather has persistent effects that independently affect suppression effort, which is why an ideal instrument is ephemeral. Valid criticisms can be made of precipitation and drought index in that these conditions not only influence treatment assignment, but also dry out fuels, affecting future fire seasons.

5.1.3 Air quality

Another condition laid out in Figure 1 is favourable smoke management conditions. Van Wagtendonk (1995) write that smoke is one of the main barriers to implementing WFU in California, Wyoming, and Montana. If air quality is already poor, managers may be unwilling to face the public backlash against even more smoke generation. Butry (2005) used distance to schools, hospitals, and nursing homes (i.e., populations sensitive to wildfire smoke) as instrumental variables in his estimation using similar reasoning; that WFU is less likely in areas closer to vulnerable people.

This paper uses daily air quality as a proxy for smoke management conditions. Fixed, geographic measures violate the exogeneity condition, as subsequent fires in the same geographic location will also be subject to these conditions. In this case, distance to vulnerable populations may also affect the effort dedicated to subsequent fires. Therefore, this estimation uses daily air quality as an instrument, which is more likely to satisfy the exogeneity condition.

5.2 Difference-in-difference

Vegetation departure can be observed over time for each fire perimeter. This means the IV strategy with control variables used to identify treatment effects on suppression cost and intensity cannot be applied here, as the instruments and covariates are collinear with incident-level (fire perimeter) fixed effects. Additionally, the identification strategy should be reformulated to take advantage of the panel structure of the data.

Rather than matching fires based on spatial overlap, this approach observes each fire perimeter in 2001, 2008, and 2012 (determined by data availability). Difference-in-difference compares the evolution of vegetation departure between treatment fire perimeters (WFU), wildfire perimeters (WF), and unburned areas within designated Wilderness. I estimate two models: one in which unburned areas are included, and one that simply compares WFU fires and WF fires before and after treatment. The reason for this is that including the unburned areas as a control group may violate the parallel trends assumption, as burned and unburned areas are not subject to the same management. After severe fires, the Forest Service often takes steps to restore the burned area to mitigate

soil erosion and sediment flows, facilitate the reintroduction of species, and so on.[4] Areas that have not burned do not receive this type of management.

6 Data and sample characteristics

6.1 Data sources

I construct a comprehensive spatial dataset of fire characteristics, conditions, and suppression costs covering the 15 westernmost states in the contiguous U.S. (Forest Service regions 1-6 excluding Hawaii) spanning the period 1990 through 2011, as no WFU is recorded after 2010. This study excludes Forest Service regions 8 (Southern states), 9 (Northeastern states), and 10 (Alaska) due to data quality concerns in region 10 and lack of implementation of the policy in regions 8 and 9. [5]

Four agencies jointly manage the National Wilderness Preservation System, and each adheres to its own practices regarding wildland fire use (Alnes, 2017). The sample is restricted to Wilderness Areas managed by the Forest Service, which keeps incident-level cost data publicly available and prioritises WFU above other fuel treatments (alternative fuel treatments are only approved under unique circumstances; Parsons & Landres, 1998). The National Fire and Aviation Management Web (FAMWEB) Database compiles individual fire reports including extensive information on costs incurred, terrain and fire characteristics, and coordinates and conditions of ignition (U.S. Forest Service, 2019a). Baylis & Boomhower (2018) note that the costs are primarily wage and equipment costs.

FAMWEB data are recorded only by point of ignition, meaning that the FAMWEB data alone cannot precisely identify spatial overlap between burned areas (Hand et al., 2016). Geospatial burn perimeter data from the Monitoring Trends in Burn Severity (MTBS) project is matched to the FAMWEB point data. The MTBS programme uses Landsat satellite data developed and managed by the U.S. Geological Survey (USGS) and the National Aeronautics and Space Administration (NASA) to map the extent of large fires (Finco et al., 2012). Treatment is determined based on spatial overlap with

[4] https://www.fs.fed.us/science-technology/fire/after-fire

[5] Both FAMWEB and MTBS report very infrequent application of WFU in FS regions 8 and 9 (four WFU fires in FAMWEB; 14 in MTBS).

previous fires. To identify area overlapped with previous burns, geospatial fire perimeters were intersected and matched using ArcGIS software. For each fire, the area within its perimeter that had been burned for fuel reduction (WFU), as well as the area that had burned unintentionally (WF) are calculated, and the characteristics of these previous fires are joined to the full dataset. The spatial overlap is illustrated with two sample Wilderness Areas in Figure 2 (Wilderness Areas are in pink, fires in yellow, overlap in blue).

Figure 4: Illustration of treatment effects: Spatial overlap



(a) Frank Church Wilderness

(b) Siskiyou Wilderness

The MTBS project records fires larger than 1000 acres (405 hectares), restricting this analysis to such fires. It is standard to restrict samples to fires that have escaped 'initial attack' to account for both simultaneity and data quality (Gebert et al., 2007; Baylis & Boomhower, 2018). The definition of a fire that has escaped initial attack is typically one greater than 300 acres (121 hectares), although this varies (Loomis et al., 2019; Hand et al., 2016; Baylis & Boomhower, 2018; Gebert et al., 2007).

It should be noted that more fires are recorded in the MTBS data than in the FAMWEB data. The matched data yields only 246 fires, while MTBS reports 648 Forest Service fires over the same period. This raises concerns regarding consistency in reporting. This data incompatibility prevents Alaskan data from being used in this analysis: FAMWEB data recorded only four fires larger than 1000 acres in Alaska during the study period, while MTBS recorded 699 such fires over the same period.

FAMWEB data is supplemented with several additional sources. Daily weather data is obtained from the Western Regional Climate Centre's Remote Automated Weather Stations (RAWS) programme, which is used by the National Interagency Fire Centre (WRCC, 2019). [6] RAWS data is taken from the nearest available station on the date of discovery. The WestWide Drought Tracker compiles monthly county-level Palmer drought severity indices (Abatzoglou et al.). Daily county-level air quality data is taken from the U.S. Environmental Protection Agency (2019). I also control for population and housing values in the wildland urban interface (WUI), decadal county-level data of which is available from the University of Wisconsin (Radeloff et al., 2017).[7] Finally, I calculate the distance to values-at-risk (towns and priority watersheds) and fire breaks (major roads and bodies of water). Shapefiles are available from the U.S. Geological Survey, Forest Service, and Census Bureau.

This analysis also addresses the causal pathway by which the treatment effect occurs (Section 2.2). Intensity-weighted area is constructed by multiplying area burned (calculated based on MTBS fire perimeters) by fire flame length (Butry, 2009). Fire intensity level (FIL) is a commonly used measure of flame length and is included in USFS incident reports and the FAMWEB database. The 6-point scale is as follows: FIL 1 represents flame length 0-2 ft.; FIL 2 is 2-4 ft.; FIL 3 is 4-6 ft.; FIL 4 is 6-8 ft.; FIL 5 is 8-12 ft; and FIL 6 is 12 ft. and higher.

For fuel availability, I turn to the Landscape Fire and Resource Management Planning Tools (LANDFIRE) programme, which is run by U.S. fire management agencies (LANDFIRE, 2001; 2008; 2012). The LANDFIRE programme supplies high-resolution (30m×30m) raster satellite data. I estimate the impact of the treatment on vegetation

[6] This is not the same weather data employed by 14582 (2019).

[7] The wildland urban interface refers to the transition from human development to wildland.

departure (VDEP), a measure of the extent to which vegetation structure differs from estimated historical conditions. The metric ranges from 0 to 100, with 100 representing the highest extent of displacement. LANDFIRE products are available for the years 2001, 2008, and 2012 (Figure 3). Due to differences in how the measure is constructed year-to-year, the administrators of the programme urge caution in conducting inter-year comparisons.

6.2 Summary statistics

Summary statistics for dependent, independent, and control variables are presented in Table 1 and broken down by treatment group: WFU, WF, and control. The sample consists of 246 fires with five having been exposed to both wildland fire use (WFU) and wildfire (WF). The control group is comprised of fires that have not been exposed to previous burns. Variable descriptions and sources are given in Table 8.

There are a few features that stand out in Table 1. First, it is clear that the extent of the analysis will be constrained by treatment group size. Only 23 fires had been exposed to fuel treatment and 37 had been exposed to previous wildfire.

There are several notable differences in the mean values between treatment groups. As expected, cost per hectare is noticeably higher in the control group than either treatment group. However, average fire size for the treatment groups is much larger than for the control. The difference between fire sizes in the treatment and control groups reiterates concerns about the endogeneity of fire size (Section 5.1). Previous studies have restricted samples to large fires (Gebert et al., 2007) or used instrumental variables for fire size (Hand et al., 2016). This estimation both restricts the study sample to large fires and employs instrumental variables. Based on these summary statistics, I estimate the effect of wildland fire use on fire intensity level alone as a robustness check.

Stars indicate variables that are log-transformed (the data presented here is original). Log transformations were based on previous literature as well as to correct heavily skewed data. In the case of suppression cost, a log transformation deflates the data without the need to determine an appropriate price index.

Table 1: Summary statistics: Dependent, independent, control variables

	(1) WFU	(2) WF	(3) Control
Cost per hectare*	164.8	385.3	1092.9
	(215.2)	(774.2)	(1981.1)
Total cost (millions)*	2.490	5.940	2.837
	(5.011)	(15.70)	(5.039)
Intensity (1-6)	3.304	3.946	3.817
	(1.743)	(1.747)	(1.842)
Intensity-weighted area*	70156.2	67119.1	17055.6
	(142542.8)	(115507.2)	(34833.9)
Slope (%)	39.57	42.97	44.74
	(19.42)	(23.30)	(23.53)
Elevation (m)	2015.4	1896.1	1788.2
	(458.5)	(533.2)	(664.7)
Aspect (0-8)	4.318	4.194	4.532
	(2.750)	(2.136)	(2.465)
Grass	0.304	0.108	0.110
	(0.470)	(0.315)	(0.314)
Brush	0	0.0270	0.0628
	(0)	(0.164)	(0.243)
Slash	0.0435	0.0270	0.0314
	(0.209)	(0.164)	(0.175)
Timber	0.652	0.730	0.743
	(0.487)	(0.450)	(0.438)
Area in WUI (%)	1.391	2.405	2.880
	(1.373)	(2.432)	(4.633)
Housing in WUI (%)	49.83	63.03	63.69
	(27.80)	(27.26)	(26.69)
Population in WUI (%)	48.74	61.68	62.63
	(28.10)	(27.62)	(27.18)
Distance to town (km)*	30.03	21.73	22.79
	(10.19)	(10.41)	(11.74)
Distance to Wilderness boundary (km)*	4.204	4.949	3.654
	(4.172)	(4.289)	(4.455)
Distance to priority watershed (km)*	38.48	46.42	37.46
	(12.81)	(49.30)	(34.67)
Wind average speed (m/s)	2.043	1.838	1.738
	(1.186)	(1.143)	(1.167)
Wind maximum speed (m/s)	13.87	10.05	10.48
	(9.621)	(3.793)	(10.22)
Air temperature	20.70	21.05	21.21
	(4.790)	(4.262)	(5.315)
Relative humidity (%)	36.70	37.03	39.47
	(12.20)	(14.88)	(16.32)
Total precipitation (mm)	0.696	0.189	0.665
	(1.663)	(0.845)	(2.150)
PDSI (-10 to +10)	-1.739	-1.027	-0.717
	(2.598)	(2.619)	(2.610)
Distance to major road (km)*	112.5	98.27	92.48
	(34.60)	(39.86)	(46.46)
Distance to body of water (km)*	6.858	7.335	6.990
	(5.413)	(5.576)	(5.543)
Fire size (ha)*	19514.7	16442.2	4129.9
	(33313.3)	(27051.2)	(7281.9)
Detection delay	2.286	2.611	1.541
	(5.386)	(5.939)	(3.351)
Wildland fire use	0.348	0.324	0.267
	(0.487)	(0.475)	(0.444)
Fuel treatment area (ha)*	210.0	909.4	0
	(356.3)	(2752.3)	(0)
N	23	37	191

mean coefficients; sd in parentheses

Summary statistics for the instrumental variables (conditions for matched fires) are given in Table 2. This table describes the conditions and characteristics of matched (treatment) fires. For the control group fires (for which there are no matching fires, as they occur on previously un-burned land), missing observations are imputed using fire season average (February through October) for the study period based on nearest available weather station. Conditions for matched fires are used as instrumental variables, as they determine treatment assignment. Weather conditions reported in Table 1 for the full data are used as control variables.

Table 2: Summary statistics: Instrumental variables

	(1) WFU	(2) WF	(3) Control
Lightning	2.652	1.892	0.283
	(2.058)	(1.468)	(0.742)
Average wind speed	1.079	1.100	1.658
	(0.604)	(0.567)	(0.995)
Maximum wind speed	6.326	6.454	16.08
	(2.916)	(3.236)	(6.694)
Air temperature	13.42	12.82	12.20
	(3.632)	(3.247)	(4.897)
Relative humidity	23.62	23.54	48.16
	(9.704)	(10.48)	(18.28)
Air Quality Index	38.51	32.85	43.15
	(37.40)	(23.74)	(29.66)
N	23	37	191

mean coefficients; sd in parentheses

As expected, lightning ignitions are higher for the WFU treatment and wind speeds are lower. The patterns in air temperature, relative humidity, and air quality are not as intuitive, as we would expect lower temperatures, as well as higher relative humidity and air quality (a higher index indicates higher levels of pollution), to be correlated with wildland fire use.

Finally, vegetation departure summary statistics are presented by treatment group and year in Table 3. The patterns illustrated here (higher vegetation departure in the control group and before fires occur) seem to support the hypothesis that fire reduces vegetation departure.

Table 3: Summary statistics: Vegetation departure

		WFU		WF	
	(1) Control	(2) Pre	(3) Post	(4) Pre	(5) Post
VDEP	57.76	41.53	41.14	46.65	44.11
	(19.47)	(10.71)	(9.096)	(10.81)	(11.06)
N	2037	48	102	482	1311

mean coefficients; sd in parentheses

7 Empirical model and methods

7.1 Suppression cost

First consider the effect of wildland fire use on suppression cost:

$$SC_{ijt} = \beta_0 + \beta_1 \sum WFU_{ijt} + \beta_2 \sum WF_{ijt} + \beta_3 X_{ijt} + \epsilon_{ijt} \tag{1}$$

where SC_{ijt} is the natural log of suppression cost per hectare for fire i in Wilderness j and time t. The unit of analysis is the wildfire i, which does not have a time component, but fires are grouped within Wilderness Areas (j), which are followed over time; making this a multilevel model. The coefficient β_1 gives the treatment effect where $\sum WFU_{ijt}$ is the number of hectares with prior WFU treatment within wildfire i's boundary. Following Parks et al. (2015) and others, I also estimate the treatment effect of previous wildfire (fires that were not managed for fuel reduction benefits but either escaped or resisted control; i.e., unplanned treatment). This parameter is not instrumented and is given by β_2 in Equation (1). The term X_{ijt} is a vector of fire characteristics and covariates described in Table 8 and summarised in Table 1. The error term ϵ_{ijt} consists of time variant, time invariant, and idiosyncratic components ($\mu_j, \omega_t, \varepsilon_{ijt}$ respectively).

The treatment assignment process is as follows (14582, 2019):

$$\sum WFU_{ijt} = \alpha_0 + \alpha_1 Z_{ijt} + \nu_{ijt} \tag{2}$$

where, due to manager involvement, the error term is correlated with the unobserved error term in (1); $corr(\nu_{ijt}, \epsilon_{ijt})$. An instrument is obtained by regressing $\sum WFU_{ijt}$ on Z_{ijt} to approximate a random treatment assignment process for the endogenous variable. Substituting the instrumented variable $\sum \hat{WFU}_{ijt}$ into the structural equation, we get:

$$SC_{ijt} = \beta_0 + \beta_1 \sum \hat{WFU}_{ijt} + \beta_2 \sum WF_{ijt} + \beta_3 X_{ijt} + \epsilon_{ijt} \tag{3}$$

The question becomes how to account for the multilevel structure of the model; i.e., that cross-sectional units of observation (fires) are grouped in possibly correlated clusters. While fires could be clustered by state, Forest Service region, National Forest, Geographic Area Coordination Centre, or ecoregion, it is most intuitive to group them by Wilderness Area. The focus of this paper is the NWPS, and management decisions are made at the Wilderness level.

This paper takes a mixed effects approach as described by Bell & Jones (2015) in which, in addition to incident-level fixed covariates, random effects account for Wilderness Area residuals. Whether to estimate a fixed effects or random effects model should be informed by the research question; i.e., whether unobserved Wilderness Area heterogeneity μ_j is correlated with the error term ϵ_{ijt}. The error term may include risk aversion, biological expertise, laws, public opinion, previous escaped fires, and the capability of nearby wildland firefighting agencies. An argument could be made that these suspected components of μ_j could be correlated with some of the regressors in the model (Tables 1, 8). However, Hausman tests indicate that random effects are strongly preferred on efficiency grounds. I report the results of the fixed effects estimation in the Appendix.

Year fixed effects are also included. A test of whether year effects are jointly equal to zero was strongly rejected, so year dummies account for time-variant unobservables ω_t.

$$SC_{ijt} = \beta_0 + \beta_1 \sum \hat{WFU}_{ijt} + \beta_2 \sum WF_{ijt} + \beta_3 X_{ijt} + \gamma_j + \delta_t \tag{4}$$

where γ_j indicates Wilderness Area random effects and δ_t indicates year fixed effects.

Hand et al. (2016) offer several compelling reasons why total suppression cost per incident is a useful dependent variable. According to Hand et al. (2016), "containment efforts are typically directed along a portion of the free-burning perimeter and not applied on a per-unit area basis" (p. 83). Practically, suppression is applied to extinguish the entire fire, not individual hectares of it. Following Hand et al. (2016), I also estimate the model from above, but with total suppression cost (TSC_{ijt}) as the dependent variable:

$$TSC_{ijt} = \beta_0 + \beta_1 \sum \hat{WFU}_{ijt} + \beta_2 \sum WF_{ijt} + \beta_3 X_{ijt} + \gamma_j + \delta_t \tag{5}$$

7.2 Mechanism

7.2.1 Intensity

The intensity-weighted area regression follows from (5), as both intensity weighted area and suppression cost are continuous, incident-level variables. Intensity weighted area (IWA_{ijt}) is the area burned multiplied by the fire intensity level (FIL):

$$IWA_{ijt} = \beta_0 + \beta_1 \sum \hat{WFU}_{ijt} + \beta_2 \sum WF_{ijt} + \beta_3 X_{ijt} + \gamma_j + \delta_t \tag{6}$$

Due to concerns regarding simultaneity between treatment and fire size, I also estimate the effect of wildland fire use on intensity alone. This estimation takes the same form, but there are important differences in implementation. Because FIL_{ijt} is an ordered categorical variable rather than a continuous variable, a random effects ordered logistic model is estimated. Two-stage least squares estimation was undertaken manually with bootstrapped standard errors.

$$FIL_{ijt} = \beta_0 + \beta_1 \sum \hat{WFU}_{ijt} + \beta_2 \sum WF_{ijt} + \beta_3 X_{ijt} + \gamma_j + \delta_t \tag{7}$$

7.2.2 Vegetation departure

Vegetation departure is the only dataset for which panel data is available, meaning a different approach is taken. Rather than using the FAMWEB-MTBS matched data, this

estimation uses the full MTBS data, which includes 648 fires rather than the matched dataset's 246. In this specification, treatment is not based on overlap between fires, but simply indicates treatment group.

$$VDEP_{it} = \alpha_i + \delta_t + \beta_1(Type_i \times Period_t) + \epsilon_{it} \tag{8}$$

where $VDEP_{it}$ is vegetation departure in fire perimeter i and year t, α_i indicates fire perimeter (or unburned area) fixed effects, and δ_t are year fixed effects. $Type_i$ indicates whether the perimeter is a WFU, WF, or unburned area; $Period_t$ indicates the treatment period (before or after the fire has occurred). ϵ_{it} is the error term.

8 Results and discussion

The main variables of interest are β_1 and β_2 from equations (4-7) and β_1 from equation (8). In the interest of concision, only these coefficients are reported. For each specification, a full set of covariates summarised and described in Tables 1 and 9 is included in the regression. [8] Also in the interest of concision, tables presented here only include results for strong instrumental variables (ignition source and maximum wind speed). Full regression results with separate IV regressions (including those of weak instruments) are reported in the Appendix. A discussion of instrument validity is undertaken after the presentation of results.

The results for all dependent variables are summarised in Table 4 with the first two columns indicating the treatment effects on per-hectare suppression costs (SC), then total suppression cost (TSC), intensity-weighted area (IWA), fire intensity level (FIL), and vegetation departure (VDEP).

[8] In the IWA specification, fire size is excluded as a control variable.

Table 4: Results: SC, TSC, IWA, FIL, VDEP

	SC		TSC		IWA		FIL		VDEP	
	(1) OLS	(2) IV	(3) OLS	(4) IV	(5) OLS	(6) IV	(7) OLS	(8) IV	(9) Total	(10) Reduced
WFU	-0.113	-0.166	-0.252	-0.710*	0.177**	0.365***	-0.040	0.121	4.188***	1.672**
	(0.093)	(0.207)	(0.203)	(0.379)	(0.073)	(0.122)	(0.102)	(0.275)	(0.744)	(0.790)
WF	-0.222**	-0.227**	-0.342	-0.369**	0.236***	0.245***	-0.013	-0.004	2.140***	-2.390***
	(0.111)	(0.091)	(0.244)	(0.167)	(0.057)	(0.054)	(0.087)	(0.128)	(0.486)	(0.807)
Constant	9.408*	7.881	3.735	-0.274	12.151***	11.813***	–	–	53.529***	46.169***
	(5.047)	(5.801)	(7.787)	(10.495)	(2.540)	(3.319)	–	–	(0.153)	(0.249)
N	213	213	213	213	213	213	213	213	3980	1943
R-sq.	0.358	0.334	0.307	0.276	0.408	0.380	–	–	0.347	0.253
First stage F statistic	–	35.06	–	35.40	–	43.47	–	30.91	–	–
Controls	YES	YES	YES	YES	YES	YES	YES	YES	NO	NO
Wilderness Area RE	YES	YES	YES	YES	YES	YES	YES	YES	NO	NO
Fire perimeter FE	NO	NO	NO	NO	NO	NO	NO	NO	YES	YES
Year FE	YES	YES	YES	YES	YES	YES	YES	YES	YES	YES
Robust (cluster) std. errs.	YES	NO	YES	NO	YES	NO	YES	NO	YES	YES
Hansen's J	–	1.195	–	6.121	–	0.421	–	–	–	–
	–	(0.274)	–	(0.013)	–	(0.517)	–	–	–	–

Standard errors in parentheses
* p<0.10, ** p<0.05, *** p<0.01

8.1 Suppression cost

In the per-hectare suppression cost specification, both the dependent and independent variables are log-transformed so the treatment effect can be interpreted as follows: A 1% increase in the number of hectares burned for fuel reduction benefits (WFU) is associated with a statistically insignificant reduction in suppression cost per hectare of between 11.3% and 16.6%, holding covariates from Table 1 constant. A 1% increase in wildfire area burned implies a statistically significant reduction in suppression cost per hectare of between 22.2% and 22.7%, all else constant. The effects of wildfire are more robust, precise, and significant perhaps due to the far higher incidence of wildfire than fuel treatment and because wildfire is not instrumented.

Following Hand et al. (2016), this paper also investigates the impact of treatment on total incident-level suppression cost. Again, the effect of wildland fire use is consistent in direction, if not in magnitude. A 1% increase in fuel treatment area predicts a reduction in total suppression cost of between 25.2% and 71.0%. A 1% increase in unplanned fuel treatment (WF) is associated with a 34.2% to 36.9% reduction in per-incident suppression cost. For both WFU and WF, IV estimates for TSC are statistically significant, while estimates from the simple random effects model are insignificant.

This lack of statistically significant fuel reduction treatment effects is consistent with findings presented by Loomis et al. (2019), Butry (2009), and Prestemon et al. (2002). Loomis et al. (2019) ascribe this to the opposing effects of fuel treatment (reducing fuels but exposing the understory to heat and wind), as well as the rarity of overlap between burned areas. However, this analysis shows that the lack of statistically significant treatment effects is likely driven by the rarity of overlap rather than the opposing effects of fuel treatment, as wildfire seems to have a consistent suppressive effect on successive fires (Prestemon et al., 2002). The mean cost per hectare for untreated fires is $1,093 per hectare. Based on the results presented in Table 4, fuel treatment could – if applied at the level of natural fire – reduce suppression costs by between $243 and $248 per hectare. The mean total per-incident cost for untreated fires was $2.837 million. Applying wildland fire use at the scale of wildfire could reduce the total suppression cost per fire by $970,000 to $1.047 million.

One caveat should be made in these interpretations of the potential of wildland fire use. A key difference between these fires is that WFU fires are not actively suppressed, while wildfires burn despite firefighters' efforts to suppress them. In the course of fighting a wildland fire, crews dig trenches, ignite backburns, carry out water or fuel drops, undertake mechanical fuel reduction, and otherwise alter the environment. It may be that these actions affect the behaviour of successive fires, in which case wildland fire use applied on a larger scale will not achieve the same results as wildland fire (14582, 2019).

I also expand upon previous analyses by modelling treatment effects over time for per-hectare and total suppression cost by estimating treatment effects for fires that occur within 2 years, 2-4 years, and 4-8 years of treatment. The longest delay between treatment and successive fire in this sample was about 8 years, so this was used as the outer bound. Fernandes & Botelho (2003) write that "fuel accumulation rate frequently limits prescribed fire effectiveness to a short post-treatment period (2-4 years)" (p. 117), so these determine the remaining two cut-offs. Due to small sample sizes, I do not separate WFU and WF fires and simply estimate the effect of fire combined. Therefore, I estimate a simple random effects model rather than the TSLS regression. The results are given in Table 5.

Table 5: Results: Treatment effect persistence

	SC			TSC		
	(1)	(2)	(3)	(4)	(5)	(6)
	<2 years	2-4 years	4-8 years	<2 years	2-4 years	4-8 years
WFU+WF	-0.256**	-0.189	-0.183*	-0.437*	-0.565	-0.266
	(0.125)	(0.252)	(0.109)	(0.252)	(0.589)	(0.247)
Constant	10.983**	11.388**	10.896**	6.055	5.907	6.098
	(5.063)	(5.113)	(5.135)	(8.085)	(7.680)	(7.777)
N	213	213	213	213	213	213
R-sq.	0.347	0.337	0.350	0.293	0.287	0.291

Standard errors in parentheses
* p<0.10, ** p<0.05, *** p<0.01

The results indicate some deterioration both in magnitude and significance. A 1% increase in burned area within the previous two years is associated with a statistically significant reduction of 25.6% in per-hectare suppression costs. This drops to 18% to

19% after 2 years. In terms of total suppression cost, only the effect of fire burned within the past two years was statistically significant (although the magnitude of the reduction associated with 2-4 year old treatment was larger).

8.2 Mechanism

8.2.1 Intensity

In addition to the research question, I explore the hypothesised mechanism. The widely held (but under-supported) consensus is that fuel treatment consumes overgrowth, thereby reducing fire intensity and ultimately reducing suppression costs. To determine the effect of fuel treatment on fire intensity, I follow Mercer et al. (2007) and Butry (2009) and estimate treatment effects on intensity-weighted area. The results for the natural log of intensity-weighted fire size are given in columns 5 and 6 of Table 4.

Table 4 indicates that fuel treatment (both wildland fire use and unplanned treatment) are strongly and positively associated with intensity-weighted area, with a 1% increase in fuel treatment associated with a 17.7% to 36.5% increase in intensity-weighted area and a 1% increase in wildfire associated with a 23.6% to 24.5% increase in intensity-weighted area. This would seem to disprove the hypothesis that previous wildfire makes future fires less intense. However, simultaneity between treatment and wildfire size is a serious concern in this estimation, and it was hoped that using an instrumental variables strategy would mitigate this. However, a subtle distinction should be made that may account for these results. We instrument for treatment assignment, not treatment exposure. In other words, this strategy is more likely to account for endogeneity in the *assignment* of treatment, not endogeneity in the *exposure* to treatment.

Due to concerns surrounding the endogeneity of fire size, I also estimate the impact of fuel treatment on fire intensity level, which is an ordered categorical variable. To estimate an ordinal logistic regression with instrumental variables, I estimate two-stage least squares manually with bootstrapped standard errors. These results provide some evidence that the intensity-weighted area estimation was indeed prone to endogeneity. While the effect of wildland fire use was not consistent across OLS and IV strategies, wildfire was associated with an almost negligible, statistically insignificant downward

impact on fire intensity level (flame length). The effect of a 1% increase in WFU on a fire's log odds of being in a higher fire intensity level is between a reduction of 0.04% and an increase of 12.1%, holding controls constant. A 1% increase in wildfire reduces the log odds of being in a higher fire intensity level between 0.004% and 0.013%. None of these estimates is statistically or practically significant. This indicates that the strong positive results for intensity-weighted area are driven endogenously by fire size rather than by intensity.

8.2.2 Vegetation departure

Finally, this analysis goes further than most economic fuel treatment evaluations by estimating the impact of treatment on fuel availability itself. I estimate the treatment effect of wildland fire use and wildland fire on vegetation departure, the extent to which an area is displaced from its estimated historical condition. Results from the difference-in-difference strategies for both WFU and WF fire are presented in columns 9 and 10 of Table 4. Column 9 gives the results of the full sample, including the vegetation departure within designated Wilderness Areas as a control group; column 10 gives results from the reduced regression where the sample is restricted to fire perimeters.

For both the full and reduced samples, these results indicate that the impact of WFU was a statistically significant *increase* in vegetation departure; i.e., that WFU led treated areas to become *more* displaced from their natural conditions by a factor of between 1.672 (reduced sample) and 4.188 (full sample) percentage points (0-100 scale). This is particularly puzzling, given the patterns in Table 3, which seem to indicate higher vegetation departure among the control group and small reductions in vegetation departure from pre-treatment to post-treatment.

The regression results for wildland fire are also perplexing. Under the full sample, WF (like WFU) as associated with a statistically significant *increase* in vegetation departure. Under the fires-only sample, WF is associated with a statistically significant *decrease* in vegetation departure of 2.390 percentage points.

One explanation for these results is that Wilderness Areas are less displaced in the first place. Wilderness fires are less threatening to lives and property, so resource constraints may have dampened the fervour of the 10 A.M. policy in Wilderness (Kilgore, 2007).

Yet, Table 3 indicates that these areas are at least somewhat displaced, with VDEP indices from about 40 to 60. Alternatively, the increase in VDEP could highlight the challenges associated with reintroducing natural fire (Section 2.1). This explanation is not necessarily undercut by the intensity results, as these indicate the intensity of subsequent fires, not the intensity of the treatment fire itself. Yet another possible justification is in temporal patterns in fuel treatment. With 2001 as the base year, 2008 exhibited consistently higher levels of departure and 2012 exhibited consistently lower levels of departure. Patterns in wildfire seasons, funding cycles, and so on could create effects that are picked up by the year fixed effects terms.

8.3 Instrument validity

Ignition source and maximum wind speed consistently perform well as instrumental variables, with suitably high first stage F statistics and first stage coefficients in line with expectations (i.e., ignition source is positively correlated with treatment; wind speed is negatively correlated). Tests of overidentifying restrictions confirm that lightning and wind speed are exogenous in the per-hectare suppression cost and intensity-weighted area specifications but indicate that they are endogenous in the total suppression cost specification. Heckman (2010) notes that the test of overidentifying restrictions is not definitive, as instruments may identify different parts of the marginal treatment curve. Due the positive result for two out of three tests, and this qualification, I tentatively accept ignition source and maximum wind speed as valid instruments for the fuel treatment variable. With that said, ignition source may be subject to some bias. As noted by Prestemon et al. (2002), there may be spatial patterns in ignition source that affect suppression cost. Additionally, it consistently gives larger treatment effects than OLS and the wind speed IV.

Average wind speed, air quality and air temperature exhibited weak predictive ability, meaning these estimates are not trustworthy. Relative humidity demonstrated reasonably high first stage F statistics, but the direction of correlation with the treatment variable was counterintuitive. These results are therefore treated with scepticism and reported in the Appendix.

The poor performance of air quality, relative humidity, and air temperature as instrumental variables may be explained by the wide range of acceptable conditions for prescribed burning. While these vary regionally and by source, Weir (2009) offers the following summary of conditions under which prescribed burning can be undertaken safely.

Table 6: Safe conditions for prescribed burning (Weir, 2009)

Condition	Minimum	Maximum
Air temperature (°C)	-1	43
Relative humidity (%)	10	80
Wind speed (m/s)	2	11

In our sample, there is not a single observation that falls outside of the safe temperature range (the highest temperature is 33 ° C). Nearly all of the observations fall within the prescribed burning range for relative humidity (only 9 are below 10%). Air quality is not typically included as a condition for prescribed burning, but the air quality in this sample was very good. Air quality index ranges from zero (good) to 500 (hazardous). The highest in this sample was 126 (unhealthy for sensitive groups). Maximum wind speed was the only weather condition IV that consistently performed as expected, and that may be because the range of acceptable conditions is decidedly narrower, with safe wind speed ranging from 2 m/s to 11 m/s.

To test the relevance of these cut-offs, I undertake the suppression cost estimations with the wind speed and relative humidity instrumental variable redefined as a dummy indicating whether conditions are safe for prescribed burning, and air quality redefined as a dummy indicating whether air quality is good or moderate (rather than unhealthy for sensitive groups). The results of these alternative IVs are given in Table 7.

Table 7: Results: Prescribed burning safe conditions cut-off test

	SC			TSC		
	(1) Wind	(2) Air quality	(3) RH	(4) Wind	(5) Air quality	(6) RH
WFU	-0.293	1.507	0.367	-0.791	1.485	-0.209
	(0.226)	(1.461)	(0.304)	(0.490)	(1.740)	(1.009)
WF	-0.244**	0.006	-0.145	-0.381	-0.022	-0.220
	(0.120)	(0.239)	(0.131)	(0.260)	(0.334)	(0.320)
Constant	7.655	9.359	-3.300	-0.348	-2.544	-17.265
	(5.160)	(7.954)	(10.018)	(8.142)	(11.924)	(17.453)
N	213	213	213	213	213	213
R-sq.	0.334	0.048	0.116	0.274	0.077	0.118
First stage F statistic	25.70	2.46	5.43	25.70	2.86	5.24

Standard errors in parentheses
* p<0.10, ** p<0.05, *** p<0.01

Under this specification, wind speed keeps its explanatory power, while the strength (weakness) of air quality as an IV is relatively unaffected. The first stage coefficient for relative humidity was in the expected direction, but still a weak predictor of WFU. This reinforces the strength of maximum wind speed as an instrumental variable.

9 Conclusion

Wildfires in the American West are becoming increasingly destructive, making it important that we identify cost effective management strategies. Wildland fire use is a compelling policy that harnesses natural processes to restore ecosystem function. In this sample, fires managed for fuel benefits cost about $1 million per incident, while suppressed fires cost more than four times as much. [9] This analysis has shown that WFU has resulted in statistically insignificant cost reductions and that WF has produced significant reductions in suppression cost. This may indicate that WFU applied on a wider scale could produce significant results (including possible co-benefits like enhanced ecosystem functioning). However, we were not able to identify the mechanism by which this occurs, as neither planned nor unplanned treatment seems to reduce fire intensity or

[9] WFU costs include monitoring and containment, as well as costs associated with risk and the need to have firefighters standing by. It is not clear whether non-monetary costs are included in these estimates and the true cost of WFU may be higher than what is reported here.

vegetation departure.

Another possible explanation is that cost reductions are achieved through learning effects. Through fighting a fire, crews may gain familiarity with an area and its conditions. When the area is re-burned, fire managers and fire crews can draw on their experiences with previous fires to inform their strategy.

9.1 Limitations

This research is subject to a number of limitations., the first of which is data quality. FAMWEB has been criticised for inconsistent reporting (Baylis & Boomhower, 2018). It also only records conditions at point of ignition, rather than conditions over the space and time the fire occupies. Parks et al. (2015) found several instances where fires had been incorrectly mapped in the MTBS data. While each dataset had its own limitations, combining them loses a substantial amount of information: 22 fires from the FAMWEB data were unmatched, as were 402 fires from the MTBS data. In addition to concerns of reporting bias, this leads to restrictions imposed by sample size (which is partly a function of data quality and partly an unavoidable consequence of the rarity of overlap between fires).

Another possible weakness has to do with using an IV regression with heterogenous causal effects. If an instrumental variable Z has variable impact on WFU across units, the IV estimator will assign greater weight to those for whom Z had a large impact on WFU. In such cases, we estimate not the average treatment effect (ATE) of the policy, but the local average treatment effect (LATE). Heckman & Urzua (2010) and Deaton (2009) contend that IV has little value, but Angrist & Imbens (1995) and Imbens (2010) argue that even if IV regression does not estimate the ATE, it still produces an estimate worthwhile for policy evaluation (e.g., average treatment effect on the treated, ATT).

This paper examines the effect of wildland fire use on suppression cost, but Wilderness management policy does not prioritise cost reductions. Interagency policy directs managers to prioritise Wilderness character over cost and convenience (Alnes, 2017). In short, we may not be studying what the policy is targeting. Additionally, Wilderness is a highly unique land designation, meaning these results are not generalisable beyond the

NWPS.

This analysis treats each fire as a single event. Fires are dynamic, complex systems that can last weeks or even months, and weather conditions, fire behaviour, and management strategies change.

Furthermore, there are two important background trends and feedback loops unaccounted for: ex-urban development and climate change, both of which exacerbate wildfire seasons and contribute to spiralling costs, and both affect fuel treatment policy.

9.2 Recommendations for further research

There are a number of avenues for further research. While this paper looked at the effect of explicit overlap, future studies should account for bordering or adjacent fires and landscape-level impacts (Parks et al., 2015). Future studies should account for the dynamic nature of wildfire, climate change, and development patterns. Additionally, it may be helpful to assess moral hazard and perverse incentives in WFU treatment assignment. For example, Vaillant & Reinhardt (2017) find that areas in the highest hazard class had the lowest percentage of area treated and the highest levels of displacement.

Finally, this topic represents a fruitful area for interdisciplinary research. This branch of economics should be better integrated with restoration ecology, fire science, and hydrology.

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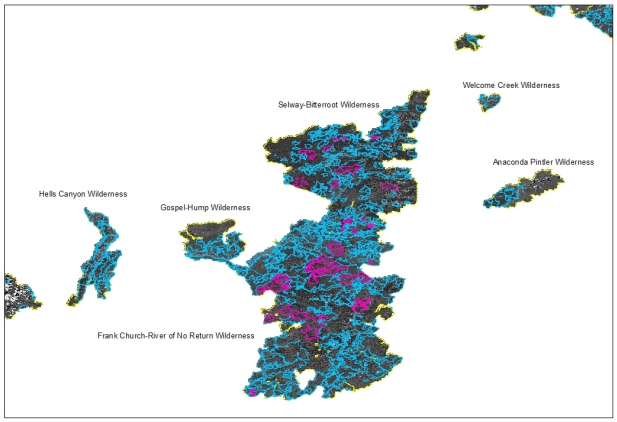
Appendices

A Variable descriptions and maps

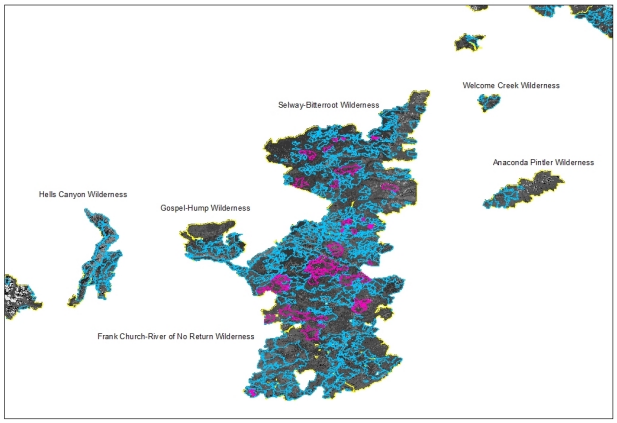
Table 8: Variable descriptions and sources

Variable	Description	Source
Dependent variables		
Marginal productivity of suppression (SC)	Cost (USD) per hectare burned	FAMWEB
Total suppression cost (TSC)	Cost (USD) per incident	FAMWEB
Fire intensity level (FIL)	Flame length (1-6 scale)	FAMWEB
Intensity weighted area (IWA)	FIL × Fire size (ha)	FAMWEB
Vegetation departure (VDEP)	0-100 scale indicating displacement from estimated historical conditions	LANDFIRE
Instrumental variables		
Ignition source	0-1 Indicating whether treatment fire was lightning-ignited	FAMWEB
Wind speed	Treatment fire maximum daily wind speed	RAWS
Air temperature	Treatment fire average daily air temperature (°C)	RAWS
Relative humidity	Treatment fire average daily humidity	RAWS
Air quality	Daily air quality index	US EPA
Control variables		
Environmental conditions		
Slope	Gradient (%) at point of ignition	FAMWEB
Elevation	Metres above sea level at ignition point	FAMWEB
Aspect	Aspect in 45°increments	FAMWEB
Fuel type	NFDRS fuel model (grass, brush, slash, timber)	FAMWEB
Values-at-risk		
WUI area	Area (km) in the wildland urban interface (WUI)	UW Silvis Lab
WUI housing	Housing units in the WUI	UW Silvis Lab
WUI population	Population in the WUI	UW Silvis Lab
Distance to town	Distance to nearest populated area (m)	U.S. Census Bureau
Distance to boundary	Distance to Wilderness Area boundary (m)	USDAFS
Meteorological conditions		
Average wind speed	Average daily wind speed (m/s)	RAWS
Maximum wind speed	Maximum daily wind speed (m/s)	RAWS
Air temperature	Average daily air temperature (°C)	RAWS
Relative humidity	Average daily humidity (%)	RAWS
Total precipitation	Total daily precipitation (mm)	RAWS
Palmer Drought Severity Index	Spans -10 (dry) to +10 (wet)	WWDT
Fire characteristics		
Fire size	Hectares burned	MTBS
Distance to road	Distance to major roadway (m)	U.S. Census Bureau
Distance to water	Distance to body of water (m)	USGS
Detection delay	Time (days) from ignition to discovery	FAMWEB

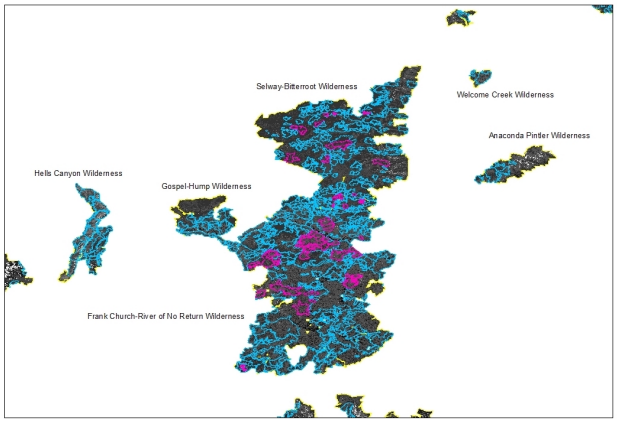
Figure 5: Vegetation departure trends in a sample Wilderness



(a) 2001



(b) 2008



(c) 2012

Blue perimeters indicate WF fire perimeters; pink represent WFU perimeters.

B Complete IV results: Random effects models

Table 9: Results: Suppression cost per hectare (SC)

	OLS	Instrumental variables					
	(1) RE	(2) Lightning	(3) Wind maximum	(4) Wind average	(5) Air quality	(6) Relative humidity	(7) Air temperature
WFU	-0.113	-0.305	-0.040	0.024	1.784	-0.242	-1.390
	(0.093)	(0.254)	(0.216)	(0.444)	(3.207)	(0.280)	(0.923)
WF	-0.222**	-0.245**	-0.212*	-0.201	-0.019	-0.237**	-0.348*
	(0.111)	(0.124)	(0.113)	(0.125)	(0.411)	(0.115)	(0.195)
Constant	9.408*	7.529	8.718	6.162	18.654	7.418	5.844
	(5.047)	(5.309)	(5.407)	(5.653)	(14.274)	(5.258)	(7.547)
N	213	213	213	213	213	213	213
R-sq.	0.358	0.332	0.337	0.283	0.057	0.329	0.218
First stage F statistic		25.30	24.40	5.48	0.36	14.44	6.71
Controls	YES	YES	YES	YES	YES	YES	YES
Wilderness Area FE	NO	NO	NO	NO	NO	NO	NO
Wilderness Area RE	YES	YES	YES	YES	YES	YES	YES
Year FE	YES	YES	YES	YES	YES	YES	YES
Robust standard errors	YES	YES	YES	YES	YES	YES	YES
LIML		NO	NO	NO	NO	NO	NO

Standard errors in parentheses

* p<0.10, ** p<0.05, *** p<0.01

Table 10: Results: Total suppression cost per incident (TSC)

	OLS	Instrumental variables					
	(1) RE	(2) Lightning	(3) Wind maximum	(4) Wind average	(5) Air quality	(6) Relative humidity	(7) Air temperature
WFU	-0.252	-1.273*	-0.159	0.089	1.566	-0.508	-2.375*
	(0.203)	(0.668)	(0.372)	(0.767)	(2.433)	(0.688)	(1.381)
WF	-0.342	-0.447	-0.306	-0.233	-0.103	-0.345	-0.571
	(0.244)	(0.316)	(0.241)	(0.231)	(0.406)	(0.245)	(0.384)
Constant	3.735	-1.172	2.103	-2.680	8.547	0.517	-4.305
	(7.787)	(8.805)	(8.176)	(9.143)	(13.085)	(8.496)	(13.020)
N	213	213	213	213	213	213	213
R-sq.	0.307	0.250	0.286	0.216	0.100	0.284	0.179
First stage F statistic		27.14	24.50	5.38	0.55	14.67	6.71
Controls	YES	YES	YES	YES	YES	YES	YES
Wilderness Area FE	NO	NO	NO	NO	NO	NO	NO
Wilderness Area RE	YES	YES	YES	YES	YES	YES	YES
Year FE	YES	YES	YES	YES	YES	YES	YES
Robust standard errors	YES	YES	YES	YES	YES	YES	YES
LIML		NO	NO	NO	NO	NO	NO

Standard errors in parentheses

* p<0.10, ** p<0.05, *** p<0.01

Table 11: Results: Intensity-weighted area (IWA)

	OLS	Instrumental variables					
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	RE	Lightning	Wind maximum	Wind average	Air quality	Relative humidity	Air temperature
WFU	0.177**	0.271***	0.477	0.415***	0.110	0.097	-0.528
	(0.073)	(0.094)	(0.298)	(0.157)	(0.580)	(0.198)	(0.503)
WF	0.236***	0.239***	0.247***	0.247***	0.216***	0.229***	0.211***
	(0.057)	(0.061)	(0.065)	(0.064)	(0.076)	(0.053)	(0.068)
Constant	12.151***	12.107***	12.888***	11.908***	8.559***	10.771***	10.135***
	(2.540)	(2.677)	(2.671)	(2.784)	(2.746)	(2.680)	(3.003)
N	213	213	213	213	213	213	213
R-sq.	0.408	0.401	0.369	0.362	0.328	0.390	0.172
First stage F statistic		32.94	27.35	8.64	1.25	15.13	3.50
Controls	YES	YES	YES	YES	YES	YES	YES
Wilderness Area FE	NO	NO	NO	NO	NO	NO	NO
Wilderness Area RE	YES	YES	YES	YES	YES	YES	YES
Year FE	YES	YES	YES	YES	YES	YES	YES
Robust standard errors	YES	YES	YES	YES	YES	YES	YES
LIML		NO	NO	NO	NO	NO	NO

Standard errors in parentheses

* p<0.10, ** p<0.05, *** p<0.01

Table 12: Results: Fire intensity level (FIL)

	(1) RE	(2) Lightning	(3) Wind maximum	(4) Wind average	(5) Air quality	(6) Relative humidity	(7) Air temperature
WFU	-0.040	0.128	0.113	-0.331	0.200	-0.059	0.396
	(0.102)	(0.686)	(0.382)	(0.821)	(6.220)	(0.824)	(1.256)
WF	-0.013	-0.004	-0.004	-0.032	0.000	-0.014	0.014
	(0.087)	(0.125)	(0.118)	(0.152)	(0.490)	(0.233)	(0.125)
N	213	213	213	213	213	213	213
R-sq.							
First stage F statistics		19.89	30.91	7.24	0.02	32.49	4.88
Controls	YES	YES	YES	YES	YES	YES	YES
Wilderness Area FE	NO	NO	NO	NO	NO	NO	NO
Wilderness Area RE	YES	YES	YES	YES	YES	YES	YES
Year FE	NO	NO	NO	NO	NO	NO	NO
Robust standard errors	YES	YES	YES	YES	YES	YES	YES
LIML		NO	NO	NO	NO	NO	NO

Standard errors in parentheses

* p<0.10, ** p<0.05, *** p<0.01

C Complete IV results: Fixed effects models

Table 13: Results: Suppression cost per hectare (SC)

	OLS	Instrumental variables					
	(1) FE	(2) Lightning	(3) Wind maximum	(4) Wind average	(5) Air quality	(6) Relative humidity	(7) Air temperature
WFU	0.012	-0.104	0.301*	0.399	-1.002	-0.016	-2.533
	(0.096)	(0.203)	(0.168)	(0.243)	(1.795)	(0.198)	(1.862)
WF	-0.201*	-0.219**	-0.156	-0.140	-0.361	-0.205*	-0.602*
	(0.116)	(0.108)	(0.120)	(0.117)	(0.299)	(0.113)	(0.311)
Constant	0.447						
	(15.207)						
N	213	165	165	165	165	165	165
R-sq.	0.469	0.464	0.440	0.417	0.112	0.469	-1.781
First stage F statistic		37.87	13.99	4.14	1.06	8.41	1.16
Controls	YES	YES	YES	YES	YES	YES	YES
Wilderness Area FE	YES	YES	YES	YES	YES	YES	YES
Wilderness Area RE	NO	NO	NO	NO	NO	NO	NO
Year FE	YES	YES	YES	YES	YES	YES	YES
Clustered standard errors	YES	YES	YES	YES	YES	YES	YES
LIML		YES	YES	YES	YES	YES	YES

Standard errors in parentheses

* p<0.10, ** p<0.05, *** p<0.01

Table 14: Results: Total suppression cost per incident (TSC)

	(1) FE	(2) Lightning	(3) Wind maximum	(4) Wind average	(5) Air quality	(6) Relative humidity	(7) Air temperature
WFU	0.027	-1.107*	0.514	0.486	-1.538	0.111	-5.738
	(0.196)	(0.662)	(0.326)	(0.464)	(2.901)	(0.524)	(4.112)
WF	-0.167	-0.346	-0.090	-0.095	-0.414	-0.154	-1.075
	(0.270)	(0.295)	(0.288)	(0.250)	(0.431)	(0.252)	(0.727)
Constant	-19.731						
	(28.486)						
N	213	165	165	165	165	165	165
R-sq.	0.473	0.342	0.449	0.452	0.223	0.472	-2.923
First stage F statistic		37.87	13.99	4.14	1.06	8.41	1.16
Controls	YES	YES	YES	YES	YES	YES	YES
Wilderness Area FE	YES	YES	YES	YES	YES	YES	YES
Wilderness Area RE	NO	NO	NO	NO	NO	NO	NO
Year FE	YES	YES	YES	YES	YES	YES	YES
Clustered standard errors	YES	YES	YES	YES	YES	YES	YES
LIML		YES	YES	YES	YES	YES	YES

Standard errors in parentheses

* p<0.10, ** p<0.05, *** p<0.01

Table 15: Results: Intensity-weighted area (IWA)

	OLS	Instrumental variables					
	(1) FE	(2) Lightning	(3) Wind maximum	(4) Wind average	(5) Air quality	(6) Relative humidity	(7) Air temperature
WFU	1.969*	5.090***	2.311**	1.913	7.842	0.071	-5.120
	(1.046)	(1.659)	(1.163)	(1.561)	(4.907)	(2.279)	(13.805)
WF	1.404*	1.655**	1.431**	1.399**	1.877***	1.251*	0.833
	(0.720)	(0.679)	(0.642)	(0.657)	(0.604)	(0.718)	(1.525)
Constant	179.777						
	(111.156)						
N	213	165	165	165	165	165	165
R-sq.	0.418	0.346	0.418	0.418	0.162	0.392	0.045
First stage F statistic		35.80	21.87	7.93	0.97	8.54	0.39
Controls	YES	YES	YES	YES	YES	YES	YES
Wilderness Area FE	YES	YES	YES	YES	YES	YES	YES
Wilderness Area RE	NO	NO	NO	NO	NO	NO	NO
Year FE	YES	YES	YES	YES	YES	YES	YES
Clustered standard errors	YES	YES	YES	YES	YES	YES	YES
LIML		YES	YES	YES	YES	YES	YES

Standard errors in parentheses
* p<0.10, ** p<0.05, *** p<0.01